### 12.2.3 Other notable effects of Air Pollution

**Smog** is a type of air pollution caused by tiny particles in the air. The word comes from a mixture of the words smoke and fog .

Today, smog generally refers to photochemical smog, which is created when sunlight reacts with nitrogen oxides and volatile organic compounds found in fossil fuel emissions from automobiles, factories, and power plants. These reactions create ground-level ozone and particulate matter, reducing visibility. Smog can make breathing more difficult, especially for people with asthma.

Smog also affects plants and animals. It damages crops as well as causes health problems in pets, farm animals and human beings. Smog has also been known to cause corrosive damage to buildings and vehicles.

**Peroxyacetyl nitrate** (PAN) is a secondary pollutant present in photochemical smog. It is thermally unstable and decomposes into peroxyethanol radicals and nitrogen dioxide gas causing eye irritation.

**Global warming**: Increase in the concentrations of greenhouse gases such as $CO_2$, methane, nitrous oxide, CFCs, and ozone causes greenhouse effect, warming of the earth, resulting in sea level rise, submerging of islands and sea shores of various parts of the world.

**Ozone depletion**: Thinning of the stratospheric ozone layer is known as ozone depletion. Such depletion causes the 'ozone hole', resulting in poor screening of the harmful UV rays and increase in incidences of skin cancer. Some of the common agents that deplete ozone are CFCs.

**Acid rain**: Acid rain is a form of precipitation that contains acidic components, such as sulphuric acid or nitric acid. It damages trees, crops and harms marine animals (coral reefs) and induces corrosion.

### 12.2.4 Control of Air Pollution

Certain measures help to remove pollutants, reduce their presence or prevent their entry into the atmosphere.

- Trees are the best remedy for urban particulate and gaseous pollution
- Forests act as carbon sinks and lungs of the planet
- Catalytic converters in vehicles help to reduce polluting gases drastically
- Diesel exhaust filters in automobiles cuts particulates
- Electrostatic precipitators reduce release of industrial pollutants
- Cost effective air pollution treatment systems like indoor plants and high performance biofilters can improve indoor air quality

The Taj Mahal, a UNESCO world heritage site, is facing deterioration and damage by industrial gases due to several industrial units around Agra. The white marble has decolorized to yellow.

### 12.2.5 Legal Protection

- The **Air** (Prevention and Control of Pollution) **Act** was enacted in 1981 and amended in 1987 for the prevention, control and abatement of Air pollution in India.
- Traffic Emissions Standards: The Government has decided to enforce Bharat Stage VI norms from 2020.
- The Green Bench and the National Green Tribunal (NGT) give judicial safeguard to environmental protection.

Steps taken by the Central and the State governments in India:

- Road traffic rationing, encourage public transport, carpooling



TN_GOVT_XII_Bio_Zoology_EM_chapter 12.indd 204 20-03-2020 10.30.24 AM